Figure 14.21 Another edge case for A*, where the algorithm returns a non-optimal solution: since the estimate for vertex v_3 is bloated, the goal is reached through a different path before even visiting v_3; once the goal is reached, the search stops and the wrong (non-optimal) path is returned.

The previous example, in fact, shows how using the wrong estimates can lead make the search unnecessary slower, which is inconvenient, but sometimes acceptable; things, however, could be much worse, because we could also find examples where A* returns a solution whose cost is not optimal: in figure 14.21, the estimate for vertex v_3 is bloated, and consequently vertex v_G is reached through a different path before even visiting v_3; remember that once the goal is reached, the search stops and so the wrong (non-optimal) path is returned.

While sometimes this can be considered acceptable, in practice, we usually we want to avoid it: and we luckily have a wave to guarantee this.

It can be proved, in fact, that A* is *complete*[12] and *optimal* when the heuristic function satisfies two conditions: it must be *admissible* and *consistent*.

1. **Admissible (aka optimistic)**: Such a heuristic never overestimates the cost to reach the goal.
2. **Consistent**: a heuristic is consistent if, given a vertex v and any of its successor u, the estimated cost for u is at most the estimated cost for v, plus the cost of